

**Hornsby Girls' High
School
Trial Examination
1999
Mathematics
2 Unit**

**Time Allowed 3 Hours
(Plus 5 Mins Reading Time)**

Directions to Candidates

- **Attempt all Questions**
- **All questions are of equal value**
- **All necessary working should be shown. Marks may be deducted for careless or badly arranged work**
- **Standard Integrals are printed on the last page**
- **Approved calculators may be used**
- **Each question must be started on a new page**

QUESTION 1.

Start a new page

Marks

- (a) Factorise : $16x^2 - 9$ 1
- (b) Convert $\frac{3\pi}{5}$ radians to degrees 1
- (c) Find a primitive of : $3 - 2x^2$ 2
- (d) When the Goods and Services Tax (GST) is introduced, some items will be exempt while others will incur a tax of 10%. The local supermarket is advertising a service,
"Find out what you will pay under a GST".
 Meg takes her trolley to Johnny, the checkout operator, and he tells her that her bill is \$285. If the GST is imposed it will rise to \$296.40.
 What percentage of Meg's bill is GST exempt? 2
- (e) Given $f(x) = 1 - x^3$, find the value of a if $f(a) = 65$. 2
- (f) Find the values of a and b if $\frac{1}{2\sqrt{3}-1} = a + b\sqrt{3}$ 2
- (g) If $\frac{a}{b} = \frac{4}{3}$ and $a + b = 28$, find the values of a and b .

QUESTION 2.

Start a new page

Marks

(a) Differentiate:

4

(i) $(x^2 + 5)^{-3}$

(ii) $x^2 \tan x$

(iii) $\frac{\log_e x}{x}$

(b) Find $\int \frac{x^2}{x^3 - 2} dx$

2

(c) Evaluate :

2

(i) $\int_0^{\pi/4} \cos 2x \, dx$

(ii) $\int_0^{\ln 3} e^x \, dx$

(d) Points $A(1, 2)$, $B(4, 7)$ and $C(3, 8)$ form a triangle.

4

Draw the triangle in your Writing Booklet.

(i) Calculate the length of AB as a surd.

(ii) Determine the equation of AB in general form.

(iii) Find the perpendicular distance of the point C from the line AB as a surd.

(iv) Hence, or otherwise, find the area of triangle ABC .

QUESTION 3.

Start a new page

Mark

- (a) If α and β are the roots of $5x^2 - 2x - 3 = 0$, find the value of:

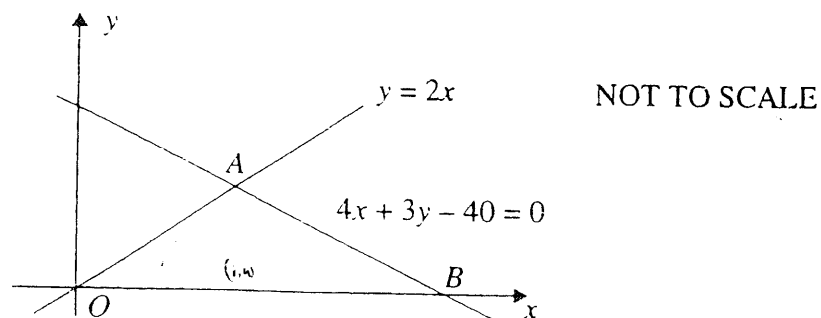
3

(i) $\alpha + \beta$.

(ii) $\frac{1}{\alpha} + \frac{1}{\beta}$.

- (b)

6



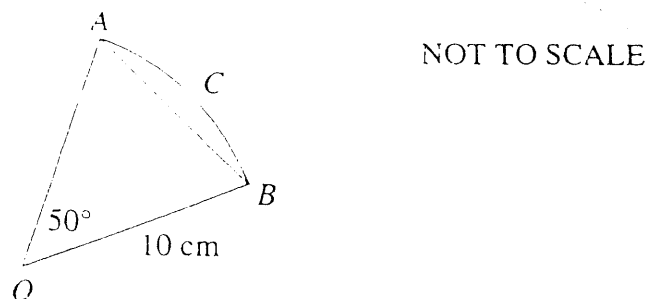
In the diagram, OA is the line $y = 2x$, and AB is the line $4x + 3y - 40 = 0$.

Copy or trace the diagram and answer the following.

- Find the coordinates of A .
- Calculate the angle of inclination of the line AB to the nearest degree.
- Write down the three inequalities satisfied by the points that lie inside $\triangle OAB$.

- (c)

3



The diagram shows a sector $OACB$ with sector angle 50° and radius 10 centimetres.

- Convert 50° to radian measure.
- Calculate, correct to one decimal place, the area of segment ABC .

QUESTION 4.

Start a new page

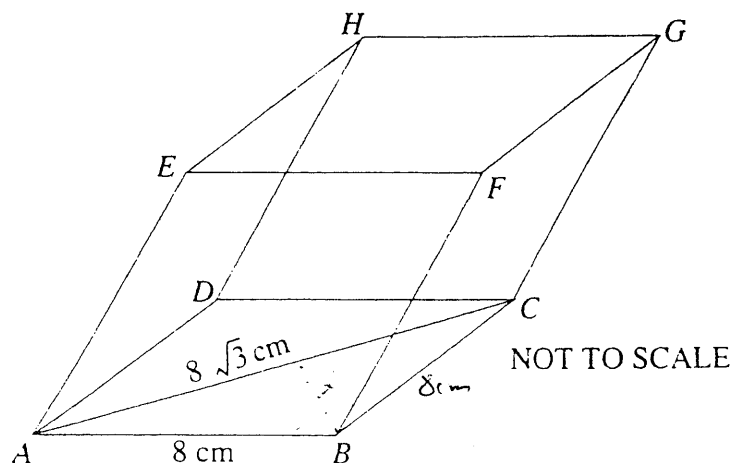
Marks

(a) Simplify $\frac{\log x^3 - \log x}{\log x^2 + \log x}$.

2

(b)

5



$ABCDEFGH$ represents a calcite crystal with six equal rhombic faces. In each rhombic face, the longer diagonal is $8\sqrt{3} \text{ cm}$, and the sides are 8 centimetres long.

- Find the size of the angle ABC in the rhombic face $ABCD$.
- Calculate the surface area of the crystal.

(c) Consider the function $y = 1 + \sin 2x$.

5

- State the period and the range of the function.
- Sketch the graph of $y = 1 + \sin 2x$ for $0 \leq x \leq 2\pi$.
- Write down the number of solutions to the equation $\sin 2x = -1$ for $0 \leq x \leq 2\pi$. (Do not solve the equation.)

QUESTION 5.**Start a new page****Mark**

- (a) Find the exact value of $\sec \frac{\pi}{4} + \cot \frac{\pi}{6}$. 2
- (b) Four metal disks, numbered 1, 2, 3 and 4 are placed in a bag. Two disks are selected at random and placed together on a tabletop to form a two-digit number. 4
- (i) By using a tree diagram, or otherwise, find how many two digit numbers which can be formed.
- (ii) Find the probability that the number formed is 21.
- (iii) Determine the probability that the number formed is divisible by 3.
- (c) Consider the arithmetic series $47 + 41 + 35 + \dots$ 6
- (i) State the common difference.
- (ii) If T_n is the n th term of the series, find the smallest n for which $T_n < 0$.
- (iii) If S_n is the sum of n terms of the series, find the smallest n for which $S_n < 0$.

QUESTION 6.**Start a new page****Marks**

- (a) A bottle of vintage wine cost \$375 when first released. After t years its value, \$ V , is given by $V = 375e^{0.05t}$. 3
- (i) Find the value of the bottle of wine after 10 years, correct to the nearest dollar.
- (ii) Find how many years it takes for the value of the wine to increase to \$1000 per bottle. Give the answer to the nearest one tenth of a year.
-
- (b) Consider the curve $y = e^{2x}(1 - x)$. 9
- (i) Find the y -intercept.
- (ii) Determine where the curve crosses the x -axis.
- (iii) Find the one stationary point, and determine its nature.
- (iv) Discuss the behaviour of the curve as $x \rightarrow +\infty$.
- (v) Discuss the behaviour of the curve as $x \rightarrow -\infty$.
- (vi) Sketch the curve.

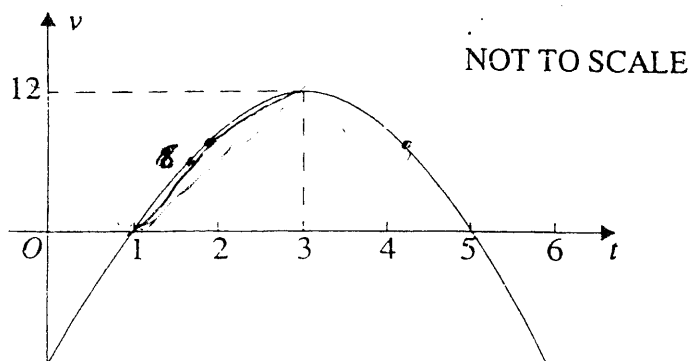
QUESTION 7.

Start a new page

- (a) A particle is moving in a straight line.

8

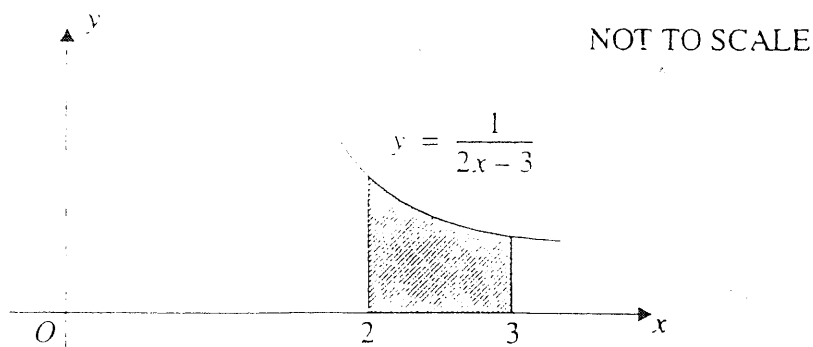
At time t seconds its velocity is v metres per second. The diagram shows the graph of v as a function of t .



- (i) At what times does the particle change direction?
- (ii) Over what period of time does the particle have positive acceleration?
- (iii) Use Simpson's rule with 3 function values to determine the distance travelled between $t = 1$ and $t = 5$.
- (iv) If the equation of the above velocity graph is $v = -3t^2 + 18t - 15$, find the distance travelled in the first 6 seconds.

- (b)

4



The diagram shows part of the graph of the function $y = \frac{1}{2x-3}$.

The shaded region is bounded by the curve, the x -axis, and the lines $x = 2$ and $x = 3$. The region is rotated around the x -axis to form a solid.

Find the exact volume of the solid in its simplest form.

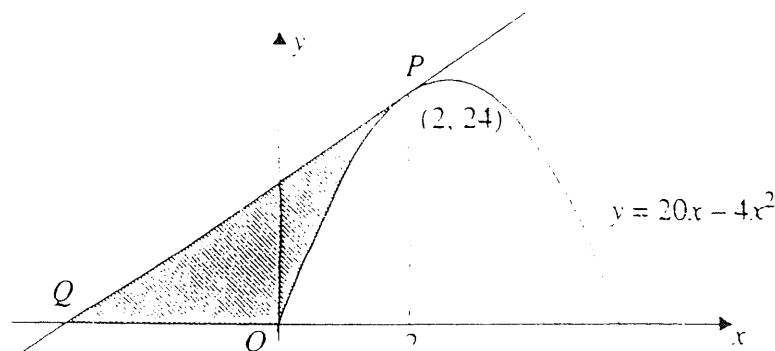
- (a) Two dice are thrown and the difference of the numbers showing is noted. 3
- (i) By using a table of outcomes, or otherwise, calculate the probability of getting a difference of four.
- (ii) If two pairs of dice are thrown what is the probability they both give a difference of four?

- (b) For the quadratic expression $mx^2 - 6x - 1$, 3

(i) find the discriminant

(ii) show that there are no values of m for which the expression is positive definite.

- (c) 6



In the graph, a tangent is drawn at $P(2, 24)$ on the parabola $y = 20x - 4x^2$. The tangent intersects the x -axis at Q .

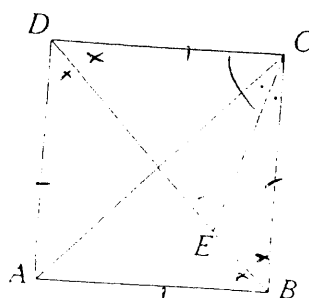
- (i) Show that the equation of the tangent is $4x - y + 16 = 0$.
- (ii) Find the coordinates of Q .
- (iii) Find the area of the shaded region POQ .

QUESTION 9.

Start a new page

- (a) If the first three terms of a geometric series are $\frac{a+b}{a-b} + m + \frac{a-b}{a+b}$, find the possible values of m .

(b)

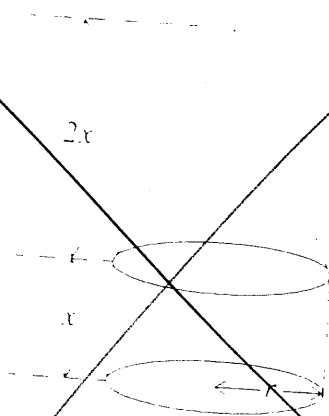


In the diagram $ABCD$ is a square with diagonals AC and BD . The point E lies on DB , and the interval CE bisects $\angle ACB$.

- (i) Prove that $\angle DCE = \angle DEC$.

(ii) Hence show $DE = DA$.

- (c) A grain silo has a cylindrical shaped wall and a cone shaped roof as in the diagram. Let the radius of the base of the silo be r metres, the height of the cylinder be x metres and the height of the cone be $2x$ metres.



- (i) Show that if the length of the slant side of the cone is 20 metres, then $r^2 = 20^2 - 4x^2$.

(ii) Show that the volume, V , of the silo is given by $V = \frac{20}{3}\pi(100x - x^3)$.

(iii) Find the exact height of the silo so that it holds the maximum amount of grain.

QUESTION 10.

Start a new page

Marks

- (a) If $9^{2n-7} = 27^{2n-5}$, find n .

2

- (b) A sum of \$2000 is invested into a retirement account and interest is compounded, at a rate of 8% per annum, every six months.

6

- (i) If no further deposit is made into the account, how much is in the account at the end of 20 years?

- (ii) Now suppose that at the beginning of the second year, and at the beginning of each subsequent year, a further \$500 is deposited into the account.

How much will be in the account at the end of 20 years?

- (c) Consider the function $f(x) = \pi x - \cos \pi x$.

4

- (i) Find $f''\left(\frac{1}{2}\right)$.

- (ii) Show that the point $\left(\frac{1}{2}, \frac{\pi}{2}\right)$ is a point of inflexion on the graph of $y = f(x)$.

20 Trial - HG HS

if $-2^{\frac{1}{2}}$

Question 1

a) $16x^2 - 9$

$= (4x-3)(4x+3)$

b) $\frac{3\pi}{5} \times \frac{180}{\pi} = 108^\circ$

c) $\int 3 - 2x^2$
 $= 3x - \frac{2x^3}{3} + C$

d) $x + y = 285$

$x + 1.1y = 296 - 40$

$y = 11.4$

$y = \$114$

$x = \$171$

$\frac{171}{285} = 60\%$ exempt.

OR
 10% = \$11.40
 of taxed
 component
 ∴ \$114 - 00
 non-taxed
 \$171

g

e) $f(x) = 1 - x^3$

$f(a) = 1 - a^3 = 65$

$-a^3 = 64$

$a^3 = 64$

$a = -4$

f) $\frac{1}{\sqrt{3}-1} \cdot \frac{2\sqrt{3}+1}{2\sqrt{3}+1}$

$= \frac{2\sqrt{3}+1}{12-1}$

$= \frac{2\sqrt{3}+1}{11} = \frac{1}{11} + \frac{2\sqrt{3}}{11}$

∴ $a = \frac{1}{11}$ $b = \frac{2}{11}$

g) $\frac{a}{b} = \frac{4}{3}$

$a = \frac{4b}{3}$

$\frac{4b}{3} + b = 28$

$4b + 3b = 84$

$7b = 84$

Question 2

a) i) $(x^2+5)^{-3}$
 $\frac{dy}{dx} = -3(x^2+5)^{-4} \cdot 2x$
 $= \frac{-6x}{(x^2+5)^4}$

ii) $y = x^2 \tan x$

$\frac{dy}{dx} = x^2 \sec^2 x + \tan x \cdot 2x$
 $= \frac{x^2}{\cos^2 x} + \frac{2x \sin x}{\cos x}$
 $= \frac{x}{\cos x} \left(\frac{x}{\cos x} + 2 \sin x \right)$

iii) $y = \frac{\log_e x}{x}$

$\frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \log_e x \cdot 1}{x^2}$
 $= \frac{1 - \log_e x}{x^2}$

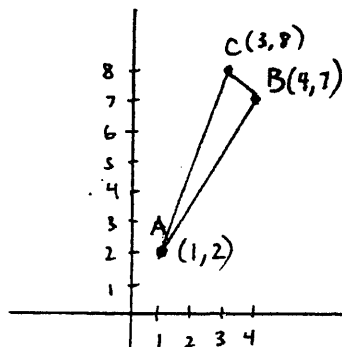
b) $\int \frac{x^2}{x^3-2} dx$

$= \frac{1}{3} \int \frac{3x^2}{x^3-2} dx$
 $= \frac{1}{3} \ln(x^3-2) + C$

c) i) $\int_0^{\frac{\pi}{4}} \cos 2x dx$
 $= \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}}$
 $= \frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{2} \sin 0$
 $= \frac{1}{2}$

ii) $\int_0^{\ln 3} e^x dx$
 $= [e^x]_0^{\ln 3}$
 $= e^{\ln 3} - e^0$
 $= 3 - 1$
 $= 2$

d)



i) $AB = \sqrt{5^2 + 1}$
 $= \sqrt{34}$

ii) $\frac{y-7}{x-4} = \frac{2-7}{1-4}$

$\frac{y-7}{x-4} = \frac{-5}{-3}$

$-3y + 21 = -5x + 2$

$5x - 3y + 1 = 0$

iii) $(3,8)$ $5x - 3y + 1 = 0$

$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|5.3 - 3.8 + 1|}{\sqrt{25 + 9}}$

$= \frac{|-8|}{\sqrt{34}} = \frac{8\sqrt{34}}{34}$

iv) $A = \frac{1}{2} \times \sqrt{34} \times \frac{4\sqrt{34}}{17} = 4 \text{ u}^2$

Question 3

a) $5x^2 - 2x - 3 = 0$

i) $\alpha + \beta = \frac{2}{5}$

ii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$
 $= \frac{\frac{2}{5}}{-\frac{3}{5}}$
 $= -\frac{2}{3}$

ii) $4x - 3y - 40 = 0$

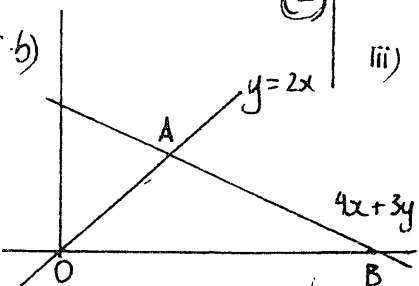
$y = 0$ $4x = 40$ $x = 10$

$B(10, 0)$

AB $m = \frac{8}{-6} = -\frac{4}{3}$

$\tan \theta = -\frac{4}{3}$

$\theta = 127^\circ$ (with + direction of x-axis)



iii) $y > 0$ $4x + 3y - 40$
 $y < 2x$

$4x + 3y - 40 = 0$

i) $y = 2x$
 $4x + 3y - 40 = 0$
 $4x + 6x - 40 = 0$
 $10x - 40 = 0$
 $10x = 40$
 $x = 4$ $y = 8$
 $A(4, 8)$

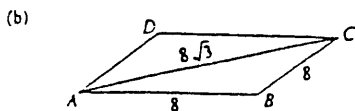
c) $50^\circ = \frac{5\pi}{18}$

ii) $A = \frac{1}{2} r^2 (\theta - \sin \theta)$
 $= \frac{1}{2} \cdot 100 \cdot \left(\frac{5\pi}{18} - \sin \frac{5\pi}{18} \right)$
 $= 5.3 \text{ cm}^2$

QUESTION 4.

$$\begin{aligned} \text{(a)} \quad \frac{\log x^3 - \log x}{\log x^2 + \log x} &= \frac{3\log x - \log x}{2\log x + \log x} \\ &= \frac{2\log x}{3\log x} \\ &= \frac{2}{3} \end{aligned}$$

Aw. 2



$$\begin{aligned} \text{(i)} \quad \cos B &= \frac{AB^2 + BC^2 - AC^2}{2(AB)(BC)} \\ &= \frac{8^2 + 8^2 - (8\sqrt{3})^2}{2 \times 8 \times 8} \\ &= \frac{64 + 64 - 192}{128} \\ &= -\frac{1}{2} \end{aligned}$$

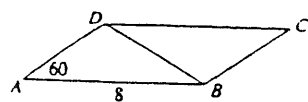
$$\therefore \angle ABC = 120^\circ$$

Aw. 2

$$\begin{aligned} \text{(ii)} \quad \text{Surface area} &= 6 \times \text{area}(ABCD) \\ &= 6 \times \left(2 \times \frac{1}{2} (AB)(BC) \sin 120^\circ \right) \\ &= 6 \times (8 \times 8 \times \sin 120^\circ) \\ &= 6 \times 8 \times 8 \times \frac{\sqrt{3}}{2} \\ &= 192\sqrt{3} \text{ cm}^2 \\ &\text{or } 333 \text{ cm}^2 \text{ (3 sig. fig.)} \end{aligned}$$

Aw. 3

Alternate Solution:



Triangle ABD is equilateral

$$\therefore BD = 8 \text{ cms.}$$

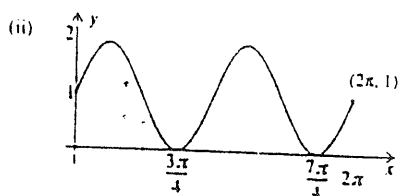
$$\begin{aligned} \therefore \text{surface area} &= 6 \times \left(\frac{1}{2} \times AC \times BD \right) \\ &= 6 \times \left(\frac{1}{2} \times 8\sqrt{3} \times 8 \right) \\ &= 192\sqrt{3} \text{ cm}^2 \end{aligned}$$

$$\text{c) } y = 1 + \sin 2x$$

$$\begin{aligned} \text{(i)} \quad \text{Period} &= \frac{2\pi}{2} \\ &= \pi \text{ units} \end{aligned}$$

$$\text{Range: } 0 \leq y \leq 2$$

Aw. 2



Aw. 2

(iii) For the graph $y = 1 + \sin 2x$ in part (ii) there are two x intercepts for $0 \leq x \leq 2\pi$.

$$\text{When } y = 0 : \sin 2x + 1 = 0$$

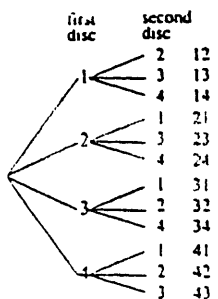
$$\therefore \sin 2x = -1 \text{ has two solutions.}$$

Aw. 1

QUESTION 5.

$$\begin{aligned} \text{(a)} \quad \sec \frac{\pi}{4} + \cot \frac{\pi}{6} &= \frac{1}{\cos \frac{\pi}{4}} + \frac{1}{\tan \frac{\pi}{6}} \\ &= \frac{1}{\frac{1}{\sqrt{2}}} + \frac{1}{\frac{1}{\sqrt{3}}} \\ &= \sqrt{2} + \sqrt{3} \end{aligned}$$

(b) (i)



Number of 2-digit numbers = 12.

Aw. 2

$$\text{(ii)} \quad P(21 \text{ is formed}) = \frac{1}{12}$$

Aw. 1

(iii) Numbers divisible by 3 are 12, 21, 24, 42

$$\begin{aligned} P(\text{number is divisible by 3}) &= \frac{4}{12} \\ &= \frac{1}{3} \end{aligned}$$

Aw. 1

$$\begin{aligned} \text{(c) (i)} \quad d &= T_2 - T_1 \\ &= 41 - 47 \\ \therefore d &= -6 \end{aligned}$$

Aw. 1

$$\begin{aligned} \text{(ii)} \quad T_n &< 0 \\ a + (n-1)d &< 0 \\ 47 + (n-1)(-6) &< 0 \\ 47 - 6n + 6 &< 0 \\ 6n &> 53 \\ n &> 8\frac{5}{6} \end{aligned}$$

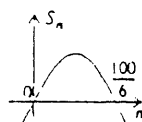
$\therefore$ smallest n is 9.

Aw. 2

$$\begin{aligned} \text{(iii)} \quad S_n &< 0 \\ \frac{n}{2} [2a + (n-1)d] &< 0 \\ \frac{n}{2} [94 - 6(n-1)] &< 0 \\ n(100 - 6n) &< 0 \\ n &< 0 \text{ No solution} \\ n &> \frac{100}{6} \end{aligned}$$

$\therefore$ smallest n is 17.

Aw. 3



QUESTION 6.

$$\text{(a)} \quad V = 375e^{0.05t}$$

$$\begin{aligned} \text{(i)} \quad V &= 375 \times e^{0.05 \times 10} \\ &= 618.27 \end{aligned}$$

$\therefore$ value is \$618 (to the nearest dollar).

Aw. 1

$$\begin{aligned} \text{(ii)} \quad 1000 &= 375e^{0.05t} \\ \frac{1000}{375} &= e^{0.05t} \\ \ln\left(\frac{1000}{375}\right) &= 0.05t \end{aligned}$$

$$t = 19.616 \dots$$

$t = 19.6$ years, correct to 1 dec. place

Aw. 2

$$\text{(b)} \quad y = e^{2x}(1-x)$$

(i) To find y -intercept, let $x = 0$

$$\begin{aligned} y &= e^{2 \times 0}(1-0) \\ &= 1 \times 1 \\ &= 1 \end{aligned}$$

$\therefore y$ intercept is 1.

Aw. 1

(ii) To find where curve crosses x -axis, let $y = 0$

$$\begin{aligned} e^{2x}(1-x) &= 0 \\ \text{Now } e^{2x} > 0 \text{ for all } x \\ \therefore 1-x &= 0 \\ x &= 1 \end{aligned}$$

$\therefore$ crosses x -axis at $x = 1$.

Aw. 1

$$\text{(iii)} \quad \frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\begin{aligned} \frac{dy}{dx} &= e^{2x} \times (-1) + (1-x) \times 2e^{2x} \\ &= -e^{2x} + 2e^{2x} - 2xe^{2x} \\ &= e^{2x} - 2xe^{2x} \\ &= e^{2x}(1-2x) \end{aligned}$$

To find stationary point, let $\frac{dy}{dx} = 0$.

$$e^{2x}(1-2x) = 0$$

$$1-2x = 0 \quad (\text{N.B. } e^{2x} \neq 0)$$

$$x = \frac{1}{2}$$

$$\text{When } x = \frac{1}{2}, y = \frac{e}{2}$$

$\therefore$ stationary point is $(\frac{1}{2}, \frac{e}{2})$.

$$\text{Now at } x = 0, \frac{dy}{dx} = e^0 \times (1-0)$$

$$= 1$$

$$> 0$$

$$\text{and at } x = 1, \frac{dy}{dx} = e^2(1-2 \times 1)$$

$$= -e^2 < 0$$

$\therefore (\frac{1}{2}, \frac{e}{2})$ is a maximum turning point ($\wedge$).

Aw. 4

$$\text{(iv)} \quad y = e^{2x}(1-x)$$

As $x \rightarrow \infty$, $e^{2x} \rightarrow \infty$ and $1-x < 0$.

$$\therefore e^{2x}(1-x) \rightarrow -\infty$$

Aw. 1

That is, the curve approaches $-\infty$.

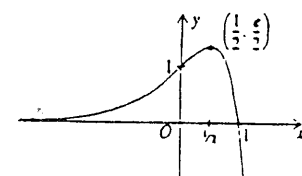
$$\text{(v)} \quad \text{As } x \rightarrow -\infty, e^{2x} \rightarrow 0 \text{ and } 1-x > 0.$$

$$\therefore e^{2x}(1-x) \rightarrow 0.$$

Aw. 1

That is, the curve approaches the x -axis.

(vi)



Aw. 1

QUESTION 7.

(a) (i) Particle changes direction at $t = 1$ and $t = 5$ seconds.

Aw.

(ii) Particle has positive acceleration for $0 < t < 3$.

Aw. 1

(iii) Simpson's Rule:

$$\begin{aligned} \text{Distance} &= \frac{b-a}{6} \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right) \\ &= \frac{5-1}{6} (0 + 4 \times 12 + 0) \\ &= \frac{4}{6} \times 48 \\ &= 32 \text{ m} \end{aligned}$$

Aw. 2

(iv) Distance travelled in first second

$$\begin{aligned} &= \left| \int_0^1 (-3t^2 + 18t - 15) dt \right| \\ &= \left| \left[-t^3 + 9t^2 - 15t \right]_0^1 \right| \\ &= \left| (-1 + 9 - 15) - 0 \right| \\ &= 7 \text{ m} \end{aligned}$$

Distance travelled between $t = 1$ and $t = 5$ is 32 m from (i).

[Note: Simpson's rule gives exact area under parabolic curves]

Distance travelled between $t = 5$ and $t = 6$ is 7 m (by symmetry)

$$\therefore \text{total distance} = 7 + 32 + 7$$

$$= 46 \text{ m}$$

Aw. 4

Alternate Solution:

Distance

$$\begin{aligned}
 &= \left| \int_0^1 (-3t^2 + 18t - 15) dt \right| + \left| \int_1^5 (-3t^2 + 18t - 15) dt \right| \\
 &\quad + \left| \int_5^6 (-3t^2 + 18t - 15) dt \right| \\
 &= \left| \left[-t^3 + 9t^2 - 15t \right]_0^1 \right| + \left| \left[-t^3 + 9t^2 - 15t \right]_1^5 \right| \\
 &\quad + \left| \left[-t^3 + 9t^2 - 15t \right]_5^6 \right| \\
 &= |(-1 + 9 - 15) - 0| \\
 &\quad + |(-125 + 225 - 75) - (-1 + 9 - 15)| \\
 &\quad + |(-216 + 324 - 90) - (-125 + 225 - 75)| \\
 &= 7 + 32 + 7 \\
 &= 46 \text{ m}
 \end{aligned}$$

(b) Volume = $\pi \int_2^6 y^2 dx$

$$\begin{aligned}
 &= \pi \int_2^6 \left(\frac{1}{2x-3} \right)^2 dx \quad \text{Aw. 1} \\
 &= \pi \int_2^6 (2x-3)^{-2} dx \\
 &= \pi \left[-\frac{1}{2} (2x-3)^{-1} \right]_2^6 \quad \text{Aw. 1} \\
 &= \pi \left[\frac{-1}{2(2x-3)} \right]_2^6 \\
 &= \pi \left[\frac{-1}{2(6-3)} - \frac{-1}{2(4-3)} \right] \quad \text{Aw. 1} \\
 &= \pi \left[-\frac{1}{6} + \frac{1}{2} \right] \\
 &= \frac{\pi}{3} \text{ cubic units} \quad \text{Aw. 1}
 \end{aligned}$$

QUESTION 8.

(a) (i) Table of differences.

		Die 1					
		1	2	3	4	5	6
Die 2	1	0	1	2	3	4	5
	2	1	0	1	2	3	4
	3	2	1	0	1	2	3
	4	3	2	1	0	1	2
	5	4	3	2	1	0	1
	6	5	4	3	2	1	0

$$\begin{aligned}
 P(\text{difference of 4}) &= \frac{4}{36} \\
 &= \frac{1}{9} \quad \text{Aw. 2}
 \end{aligned}$$

(ii) $P(\text{both give difference of 4}) = \frac{1}{9} \times \frac{1}{9}$

$$= \frac{1}{81} \quad \text{Aw. 1}$$

(b) (i) $mx^2 - 6x - 1$

Discriminant = Δ

$$\begin{aligned}
 &= b^2 - 4ac \\
 &= (-6)^2 - 4 \times m \times (-1) \\
 &= 36 + 4m \quad \text{Aw. 1}
 \end{aligned}$$

(ii) For expression to be positive definite, we require $m > 0$ and $\Delta < 0$.

$$\begin{aligned}
 \Delta &= 36 + 4m < 0 \\
 4m &< -36 \\
 m &< -9 \\
 \therefore \text{if } \Delta < 0, \text{ then } m < -9
 \end{aligned}$$

(c) (i) $y = 20x - 4x^2$

$$\frac{dy}{dx} = 20 - 8x$$

At $x = 2$, $\frac{dy}{dx} = 20 - 8 \times 2$

$$= 4$$

Equation of tangent given by

$$y - y_1 = m(x - x_1)$$

$$y - 24 = 4(x - 2)$$

$$y - 24 = 4x - 8$$

$$4x - y + 16 = 0$$

Aw. 1

QUESTION 9.

(a) (i) Geometric series is $\frac{a+b}{a-b} + m + \frac{a-b}{a+b}$

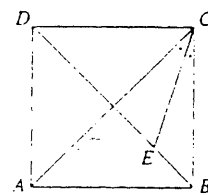
$$\frac{m}{\frac{a+b}{a-b}} = \frac{\frac{a-b}{a+b}}{m}$$

$$\begin{aligned}
 m^2 &= \frac{a+b}{a-b} \times \frac{a-b}{a+b} \\
 &= 1
 \end{aligned}$$

$$m = \pm 1$$

Aw. 2

(b)



(i) Aim to prove $\angle DCE = \angle DEC$

$ABCD$ is a square (given)

$$\therefore \angle DCA = \angle ACB$$

$$= 45^\circ \text{ (Diagonal } AC \text{ bisects } \angle DCB)$$

$$\therefore \angle ACE = 22\frac{1}{2}^\circ \text{ (Given } CE \text{ bisects } \angle ACB)$$

$$\therefore \angle DCE = 77\frac{1}{2}^\circ \text{ (} \angle DCA + \angle ACE)$$

$$\angle DEC = \angle DBC + \angle BCE \text{ (in any triangle, exterior angle = sum of interior opposites)}$$

$$= 45^\circ + 22\frac{1}{2}^\circ \text{ (same reason as above)}$$

$$= 77\frac{1}{2}^\circ$$

$$\therefore \angle DCE = \angle DEC$$

Aw. 3

(ii) From (i), $\triangle DEC$ is isosceles

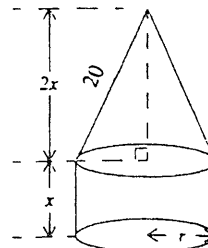
$$\therefore DC = DE \text{ (sides opposite equal } \angle s \text{ } DCE, DEC)$$

$$\text{But } DC = DA \text{ (sides of square)}$$

$$\therefore DE = DA$$

Aw. 1

(c)



(i) From diagram

$$r^2 + (2x)^2 = 20^2$$

$$r^2 + 4x^2 = 20^2$$

$$r^2 = 20^2 - 4x^2$$

Aw. 1

$$\begin{aligned}\therefore \text{volume of silo: } V &= \pi r^2 x + \frac{1}{3} \pi r^2 (2x) \\ &= \frac{5}{3} \pi r^2 x \\ &= \frac{5}{3} \pi x (400 - 4x^2) \\ &= \frac{20}{3} \pi (100x - x^3)\end{aligned}$$

Aw. 1

$$\begin{aligned}\text{(iii)} \quad \frac{dV}{dx} &= \frac{20}{3} \pi (100 - 3x^2) \\ \frac{d^2V}{dx^2} &= \frac{20}{3} \pi (-6x) \\ &= -40\pi x\end{aligned}$$

To find stationary point, let $\frac{dV}{dx} = 0$.

$$\begin{aligned}\frac{20\pi}{3} (100 - 3x^2) &= 0 \\ 3x^2 &= 100 \\ x^2 &= \frac{100}{3} \\ \therefore x &= \frac{10}{\sqrt{3}} \quad (x > 0)\end{aligned}$$

$$\frac{d^2V}{dx^2} < 0 \text{ for } x = \frac{10}{\sqrt{3}}$$

$\therefore$ maximum volume occurs where $x = \frac{10}{\sqrt{3}}$

$$\begin{aligned}\therefore \text{required height} &= 3 \times \frac{10}{\sqrt{3}} \\ &= 10\sqrt{3} \text{ metres} \quad \text{Aw. 4}\end{aligned}$$

QUESTION 10.

$$\text{(a)} \quad 92n - 7 = 272n - 5$$

$$(3^2)^{2n-7} = (3^3)^{2n-5}$$

$$3^{4n-14} = 3^{6n-15}$$

$$4n - 14 = 6n - 15$$

$$1 = 2n$$

$$n = \frac{1}{2}$$

Aw. 2

$$\text{(b) (i)} \quad A = P \left(1 + \frac{r}{100} \right)^n$$

$$= 2000 \left(1 + \frac{4}{100} \right)^{40}$$

$$= 2000(1.04)^{40}$$

$$= \$9602.04$$

Aw. 1

Aw. 1

$$\text{(ii)} \quad \text{Amount} = 500 \times 1.04^{18} + 500 \times 1.04^{16} + 500 \times 1.04^{14} + \dots + 500 \times 1.04^2$$

Aw. 1

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\therefore \text{amount} = \frac{500 \times 1.04^2 ((1.04^2)^{19} - 1)}{(1.04^2) - 1} \quad \text{Aw. 1}$$

$$= \$22790.57 \quad \text{Aw. 1}$$

$$\therefore \text{amount in account at end of 20 years} = \$9602.04 + \$22790.57$$

$$= \$32392.61 \quad \text{Aw. 1}$$

$$\begin{aligned}f''(0.4) &= \pi^2 \cos(0.4\pi) \\ &\approx 3.049 \dots \\ &> 0\end{aligned}$$

$$\begin{aligned}f''(0.6) &= \pi^2 \cos(0.6\pi) \\ &\approx -3.049 \dots \\ &< 0\end{aligned}$$

$\therefore \left(\frac{1}{2}, \frac{\pi}{2} \right)$ is a point of inflection on $y = f(x)$.

$$\text{(c)} \quad f(x) = \pi x - \cos(\pi x)$$

$$\text{(i)} \quad f'(x) = \pi + \pi \sin(\pi x)$$

$$f''(x) = \pi^2 \cos(\pi x)$$

$$f''\left(\frac{1}{2}\right) = \pi^2 \cos\left(\frac{\pi}{2}\right)$$

$$= 0$$

Aw. 1

$$\text{(ii)} \quad f\left(\frac{1}{2}\right) = \frac{\pi}{2} - \cos\frac{\pi}{2}$$

$$= \frac{\pi}{2} - 0$$

$$= \frac{\pi}{2}$$

$\therefore \left(\frac{1}{2}, \frac{\pi}{2} \right)$ lies on $y = f(x)$.

Aw. 1

$$\text{From (i), } f''\left(\frac{1}{2}\right) = 0$$

If δ is some small positive value

$$f''\left(\frac{1}{2} - \delta\right) = \pi^2 \cos\left[\pi\left(\frac{1}{2} - \delta\right)\right]$$

$$= \pi^2 \cos\left(\frac{\pi}{2} - \pi\delta\right)$$

$$> 0 \quad \left(\frac{\pi}{2} - \pi\delta \text{ is in 1st quadrant} \right)$$

Aw. 1

$$f''\left(\frac{1}{2} + \delta\right) = \pi^2 \cos\left[\pi\left(\frac{1}{2} + \delta\right)\right]$$

$$= \pi^2 \cos\left(\frac{\pi}{2} + \pi\delta\right)$$

$$< 0 \quad \left(\frac{\pi}{2} + \pi\delta \text{ is in 2nd quadrant} \right)$$

$\therefore$ concavity changes about $x = \frac{1}{2}$. Aw. 1

$\therefore \left(\frac{1}{2}, \frac{\pi}{2} \right)$ is a point of inflection on $y = f(x)$.

~~Q8~~ (ii) To find Q, let $y = 0$ in equation of tangent.

$$4x - 0 + 16 = 0$$

$$x = -4$$

$\therefore$ coordinates of Q are $(-4, 0)$

Aw. 1

$$\text{(iii)} \quad \text{Area of triangle} = \frac{1}{2} \times 6$$

$$= 72 \text{ square units}$$

$$\text{Area under curve} = \int_0^2 (20x - 4x^2) dx$$

$$= \left[10x^2 - \frac{4}{3}x^3 \right]_0^2$$

$$= \left(40 - \frac{32}{3} \right) - 0$$

$$= 29\frac{1}{3} \text{ square units}$$

$$\therefore \text{area of region } POQ = 72 - 29\frac{1}{3}$$

$$= 42\frac{2}{3} \text{ square units}$$

Aw. 4